Entanglement Dynamics in a Randomly-Kicked Ising Model

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Overview

- Kicked Ising Model
- Entanglement Entropy
- Jordan Wigner Transformations
- Random Protocol
- Bloch-Sphere Dynamics

Kicked Ising Model

The Hamiltonian

$$H(t) = \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x + \sum_{k=-\infty}^{\infty} \delta(k - \frac{t}{T}) \sum_{j=1}^{L} \sigma_j^z \tag{1}$$

The Floquet operator

$$U = \exp(-iT \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) \exp(-iT \sum_{j=1}^{L} \sigma_j^z) \tag{2}$$

When T $= \pi/4$; Entanglement Entropy

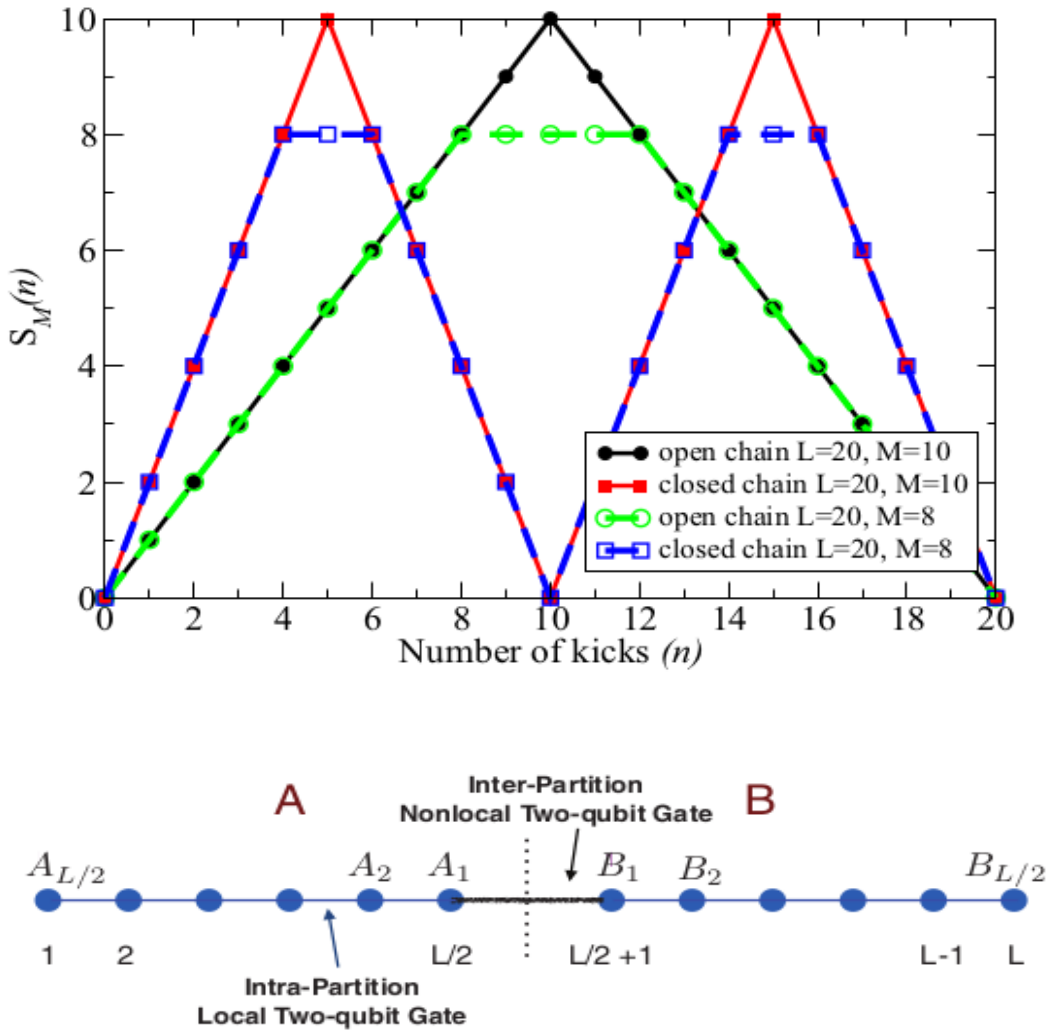


Figure: Taken from *Arul,et.al., Phys. Rev. A 91, 022318 (2015)*

Jordan Wigner Transformations

Express the system in Fermion-momentum basis:

$$H_{kick} = 2\Sigma_k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] \tag{3}$$

$$H_{interaction} = -2\Sigma_k (\cos k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] - \sin k [d_{-k} d_k + d_k^\dagger d_{-k}^\dagger]) \tag{4}$$

Define a new space whose basis states are $|0\rangle$ and $d_k^\dagger d_{-k}^\dagger |0\rangle$.

$$H_{kick} = -2\Sigma_k \sigma_k^z \tag{5}$$

$$H_{interaction} = 2\Sigma_k (\cos k \sigma_k^z - \sin k \sigma_k^x) \tag{6}$$

Floquet Operator Now

$$U = U_{interaction} U_{kick} \tag{7}$$

$$= \exp(-2iT\Sigma_k(\cos k\sigma_k^z - \sin k\sigma_k^x)) \exp(2iT\Sigma_k \sigma_k^z) \tag{8}$$

Very Simple Floquet operator!

When $T = \pi/4$,

$$U = -\prod_{\otimes k}(\cos k + i \sin k \sigma_k^y) \tag{9}$$

Random protocol

Studied by *Amit Dutta. et.al., PhysRevB.98.064305 (2018)*

$$U_n = U_{\text{interaction}}(U_{\text{kick}})^{\alpha_n}, \qquad \alpha_n \epsilon \{0, 1\}$$

"It is demonstrated that the model heats up to the infinite temperature ensemble for a minimal aperiodicity in driving."

Entanglement entropy

Von Neumann entropy:

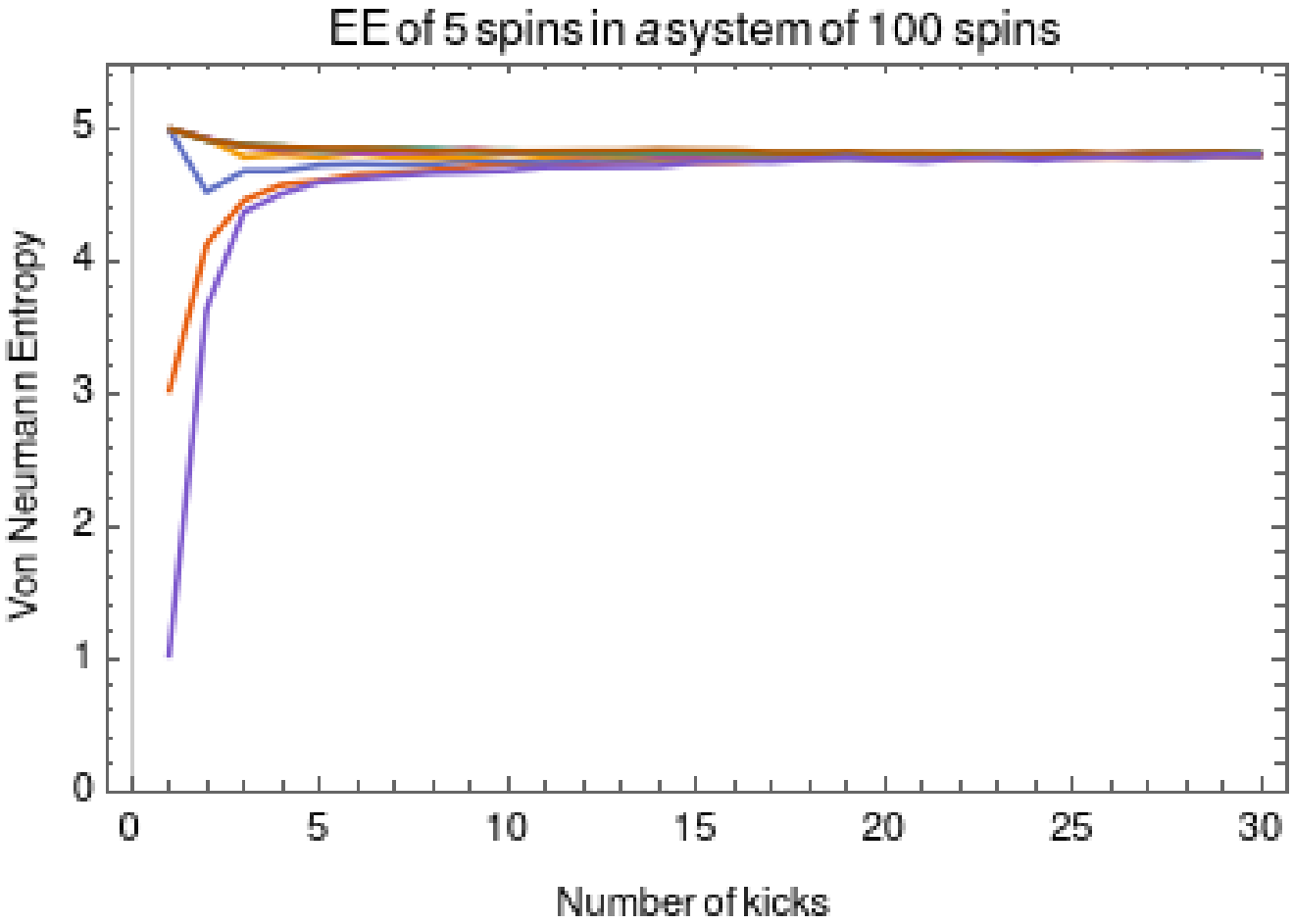
$$\mathcal{S} = -\rho \log \rho \tag{10}$$

This is equivalent to the entropy of the eigen values of the correlation matrix given by (*I.Peschel et.al., J. Phys. A: Math. Theor. 42 (2009) 504003*):

$$\mathcal{C} = \begin{bmatrix} I - C & F \\ F^\dagger & C \end{bmatrix} \text{ where, } \quad C_{mn} = \langle c_m^\dagger c_n \rangle \text{ and } F_{mn} = \langle c_m^\dagger c_n^\dagger \rangle \tag{11}$$

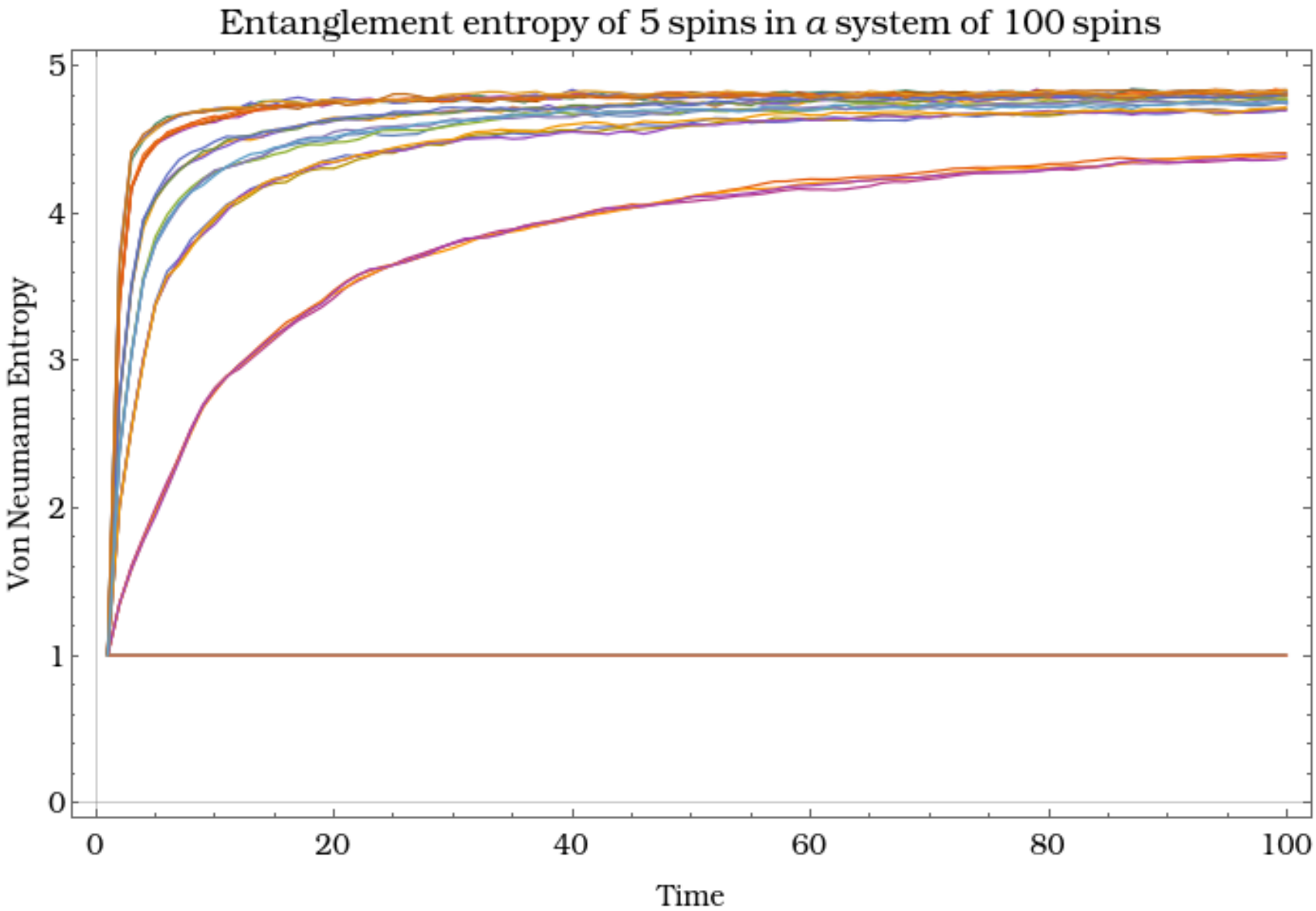
Entanglement entropy

Random case



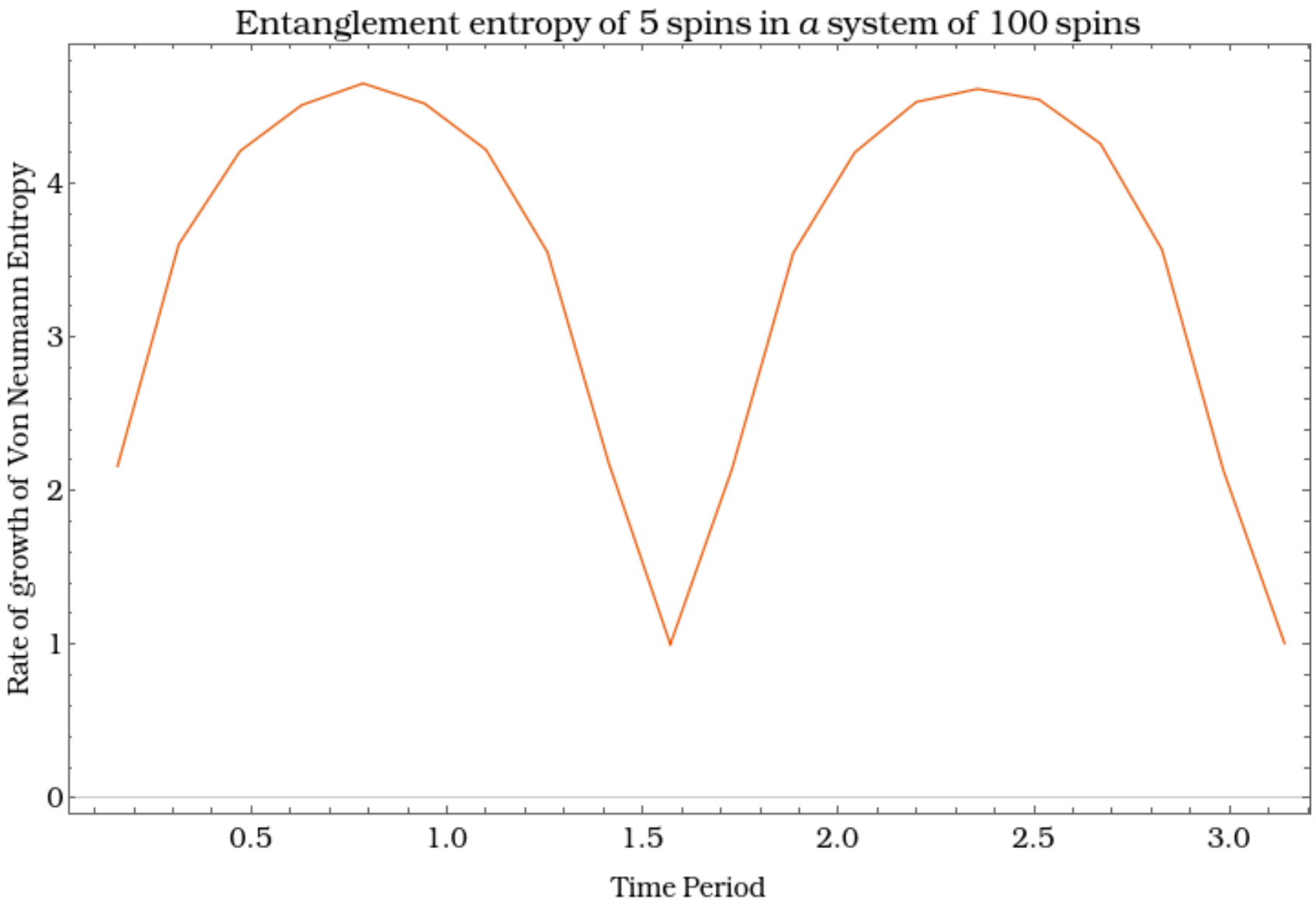
Entanglement entropy

Random case



Entanglement entropy

Random case



Evolution on the Bloch sphere

Consider a single k-mode:

$$\begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix} = e^{i\beta} \cos[\frac{\theta}{2}]|0\rangle + e^{i\alpha} \sin[\frac{\theta}{2}]|1\rangle \tag{12}$$

$$= e^{i\beta}(\cos[\frac{\theta}{2}]|0\rangle + e^{i(\alpha-\beta)} \sin[\frac{\theta}{2}]|1\rangle) \tag{13}$$

Evolution on the Bloch sphere, $T = \pi/4$

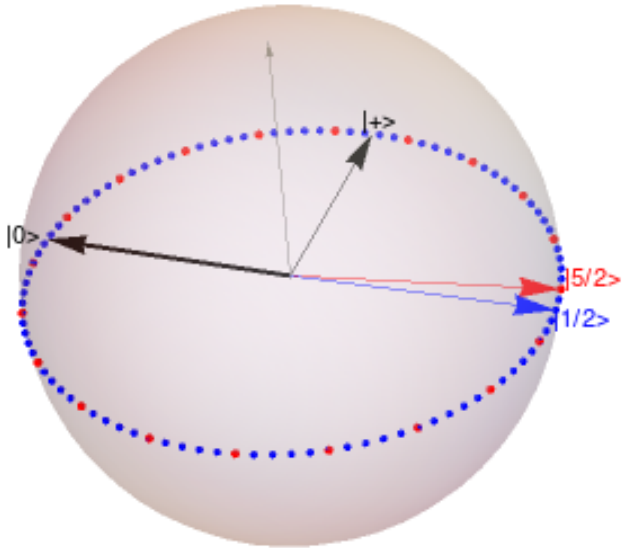


Figure: k-mode evolution, when $T = \pi/4$. The colored arrow shows the initial states for two modes $k = \pi/100$ and $k = 5\pi/100$

Evolution on the Bloch sphere, $T = \pi/4 + 0.01$

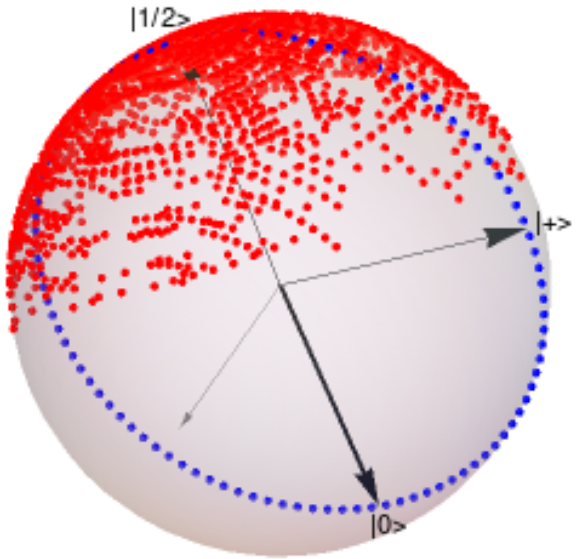


Figure: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = \pi/100$ and the blue points correspond to $T = \pi/4$

Evolution on the Bloch sphere, $T = \pi/4 + 0.01$

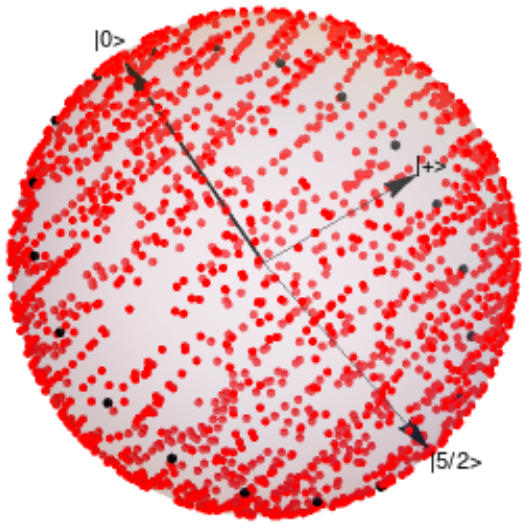


Figure: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = 5\pi/100$ and the blue points correspond to $T = \pi/4$

Remarks

- $T = \pi/4$ is seen to resemble a point of unstable equilibrium.
- There is a direct correlation between rate of growth of entanglement entropy and the extent of thermalization
- Further studies with Thue Morse/Fibonacci sequenced protocols can show interesting results

Thank you!
Questions and comments?